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AD3811 – PROJECT WORK
FIRST REVIEW

A Multi-Agent Reinforcement Learning Architecture with Independent Sentiment-Anomaly Veto for Pump-and-Dump-Resistant Trading

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Abstract

Automated trading systems built on price and volume signals cannot distinguish a genuine, fundamentals-driven price spike from a coordinated pump-and-dump, since both surface identically as "price moving up fast." Recent multi-agent reinforcement learning (MARL) architectures for trading and portfolio management — including Agentic AI ensembles that combine LSTM, Transformer, and XGBoost prediction with a FinBERT-based Sentiment Agent resolved by majority vote, and modular systems such as MSPM, which route FinBERT sentiment into asset-specific Deep Q-Network agents whose outputs feed a PPO-based portfolio allocator — already incorporate sentiment analysis. In every such system, however, the sentiment signal is treated as one more input feature blended into the decision with no explicit way to distinguish routine sentiment from statistically abnormal hype, and neither system is evaluated against labeled manipulation events.

This project proposes a Multi-Agent Reinforcement Learning Architecture with an Independent Sentiment-Anomaly Veto for pump-and-dump-resistant trading. The system comprises an Analysis Agent that proposes buy, sell, or hold actions from price-action and technical indicators via a reinforcement-learning policy; a Risk Agent that filters proposals against position-sizing and volatility limits; and a Sentiment/Hype Deflator Agent that scores live news and social media with FinBERT and flags statistically abnormal emotional intensity relative to an asset's historical baseline. Unlike prior work, the Deflator Agent sits outside the voting loop with explicit override authority to force a hedge or reversal whenever hype is anomalous, regardless of what the other agents vote.

The system will be backtested on historical price, news, and social data spanning documented pump-and-dump events, comparing a price-only baseline, a sentiment-as-feature configuration replicating existing literature, and the proposed sentiment-as-veto configuration, evaluated on Sharpe ratio, maximum drawdown, and manipulation-detection precision and recall.

Literature Survey

Recent research in artificial intelligence–driven financial trading has increasingly moved beyond isolated prediction models toward **multi-agent and agentic architectures**, in which different agents are assigned specialized analytical responsibilities. This evolution is motivated by the inherently heterogeneous nature of financial markets, where price movements are influenced simultaneously by technical indicators, fundamental information, financial news, market sentiment, macroeconomic conditions, and risk factors. Rather than relying on a single model to process all these information sources, recent studies have explored the decomposition of trading into specialized analytical and decision-making agents.

Singh [1] presents an **Agentic AI framework for stock-market prediction** that combines multiple specialized agents with machine-learning and reinforcement-learning components. The framework consists of a Market Agent for price momentum and volatility, a FinBERT-based Sentiment Agent for financial news, a Prediction Agent based on LSTM models, and a Decision Agent that resolves the outputs through majority voting. The system further employs a dynamic ensemble of LSTM, XGBoost, and Transformer models and subsequently refines decisions using Q-learning. Experiments on nine years of AAPL data demonstrate improved predictive performance, with the ensemble achieving an R^2 of approximately 0.96 and a positive Sharpe ratio. Agentic_AI_for_Intelligent_Stoc… However, the architecture treats sentiment as one component of a common decision process. Consequently, the sentiment agent does not possess independent authority to prevent a trade even when abnormal sentiment conditions indicate potential market manipulation.

Huang and Tanaka [2] approach multi-agent trading from the perspective of **scalability and portfolio management**. Their Modularized and Scalable Multi-Agent Reinforcement Learning-based Portfolio Management system (MSPM) consists of two principal components: an Evolving Agent Module (EAM) and a Strategic Agent Module (SAM). The EAM uses a DQN-based agent to process heterogeneous information and generate asset-specific signals, while the SAM employs PPO to integrate the signals from multiple EAMs and perform portfolio reallocation. The modular architecture enables trained EAMs to be reused across different portfolios. Experiments using eight years of U.S. stock-market data demonstrate substantial improvements over several baselines in accumulated return and Sortino ratio. Importantly, the authors conduct an ablation experiment comparing EAM-enabled and EAM-disabled systems, demonstrating the importance of specialized information-generating agents. 2102.03502v4 Nevertheless, the signals generated by the EAMs ultimately become information available to the portfolio-level decision agent; no specialized agent is given an explicit **hard authority to block or override a trading action**.

The development of LLM-based trading systems has further expanded the concept of specialized financial agents. Xiao et al. [3] introduce **TradingAgents**, a multi-agent LLM framework designed to emulate the organizational structure of a real trading firm. The framework incorporates fundamental, sentiment, and technical analysts, followed by Bull and Bear research agents, trader agents, a dedicated risk-management team, and a fund manager. Agents communicate through structured interactions and debate before a final trading decision is produced. The framework reports improvements in cumulative returns, Sharpe ratio, and maximum drawdown relative to baseline approaches. 2412.20138v7 The dedicated risk-management team is particularly relevant to the proposed research because it introduces a separate risk-oriented layer rather than treating all analytical signals identically. However, its role remains **risk assessment and feedback within the collaborative decision process**, rather than an independently evaluated veto mechanism capable of unconditionally blocking a trade.

Miyazaki et al. [4] extend this multi-agent paradigm by focusing on **fine-grained task decomposition and realistic investment workflows**. Their *Toward Expert Investment Teams* framework organizes the trading process hierarchically using Quantitative, Qualitative, News, Technical, Sector, Macro, and Portfolio

Manager agents. Rather than giving agents broad instructions, the authors decompose investment analysis into concrete tasks corresponding to real-world analyst workflows. Their experiments on Japanese equity data demonstrate that fine-grained task decomposition improves risk-adjusted performance, while systematic agent ablation provides insight into the contribution of individual agents. 2602.23330v1 The framework also emphasizes interpretability by examining intermediate agent outputs. This provides an important methodological foundation for the proposed system: **specialized agents should have clearly defined responsibilities and their individual contributions should be evaluated through ablation studies**. However, the final decision remains within the hierarchical manager architecture rather than being subject to an independently enforced safety constraint.

Another important direction is the integration of **reinforcement learning with financial reasoning and risk-sensitive decision making**. Benhenda [5], through FinRL-DeepSeek, investigates the use of an LLM-derived financial risk assessment signal together with risk-sensitive reinforcement learning. The approach incorporates financial-news-derived risk information into a CVaR-oriented PPO framework, thereby extending conventional return-oriented trading policies with explicit risk information. This work is particularly relevant because it demonstrates that a separate risk-oriented information channel can complement the primary trading policy. However, the risk assessment remains an **input to the reinforcement-learning policy**, rather than an independently trained mechanism that can override the policy's action.

Xiao et al. [6] further explore financial reasoning through **Trading-R1**, a financial reasoning model trained using supervised fine-tuning and reinforcement learning. The model is trained on a large heterogeneous dataset containing technical data, fundamentals, news, sentiment, and macroeconomic information. Its three-stage training process progressively develops structured reasoning, evidence-grounded analysis, and market-aware trading decisions. The model uses volatility-adjusted labels and evaluates trading performance through cumulative return, Sharpe ratio, hit rate, and maximum drawdown. 2509.11420v1 Trading-R1 demonstrates that structured reasoning and reinforcement learning can produce interpretable, risk-aware financial decisions. Nevertheless, the different information modalities are ultimately integrated into a unified reasoning and decision process rather than being separated into an independently authoritative **market-safety channel**.

A complementary approach is presented by Xiong et al. [7] through **QuantHarness**, a price-driven multi-agent LLM framework designed specifically for high-frequency trading. Unlike systems that depend heavily on news and social-media information, QuantHarness operates primarily on structured OHLC and technical-market data. It decomposes short-horizon trading into Indicator, Pattern, Trend, and Risk agents, with the Risk Agent integrating upstream signals into a risk–reward assessment. The framework emphasizes low-latency decision making, tool-grounded reasoning, and interpretable trading decisions, and reports improved directional performance across multiple financial instruments and time horizons. 2509.09995v4 This work is particularly significant for the proposed architecture because it demonstrates that **risk can be assigned to a dedicated agent rather than being treated merely as another numerical feature**. However, the Risk Agent in QuantHarness remains part of the aggregated decision pipeline; the framework does not specifically investigate whether a risk/anomaly agent should possess unilateral authority to stop a profitable-looking trade.

The above studies primarily investigate how specialized agents can improve prediction, reasoning, portfolio allocation, and risk management. A separate research direction focuses on the **detection of cryptocurrency market manipulation**, particularly pump-and-dump (P&D) schemes. La Morgia et al. [8] provide an important empirical foundation by constructing a dataset of real cryptocurrency pump-and-dump events associated with Telegram coordination groups and corresponding Binance trading activity. Their work enables manipulation detection to be evaluated using **real, documented events rather than artificially**

injected anomalies. This is particularly valuable for the proposed research because the objective is not merely to determine whether an agent can classify sentiment as positive or negative, but whether it can recognize abnormal promotional behavior that precedes or accompanies a manipulation event.

Ardia and Bluteau [9] further establish the relationship between social-media promotion and cryptocurrency price dynamics through their empirical study of Twitter and cryptocurrency pump-and-dump events. Their analysis demonstrates that coordinated promotional activity around such events has a measurable relationship with cryptocurrency returns. This provides an important financial justification for incorporating social-media/news sentiment into a trading-safety mechanism: a strongly positive sentiment signal can sometimes represent **manipulative market activity rather than genuine positive information**.

More recent work directly investigates machine-learning-based P&D detection. Yu et al. [10], in *Improving Cryptocurrency Pump-and-Dump Detection through Ensemble-Based Models and Synthetic Oversampling Techniques*, address the severe class imbalance inherent in manipulation datasets. Using the publicly available La Morgia et al. dataset, the study evaluates Random Forest, AdaBoost, GBM, XGBoost, and LightGBM with and without SMOTE oversampling. The authors report that SMOTE substantially improves detection capability, with XGBoost and LightGBM achieving recall rates of 94.87% and 93.59%, respectively. 2510.00836v1 This work demonstrates that P&D detection can be formulated as a dedicated machine-learning problem and that high recall is particularly important because manipulation events are rare but potentially highly damaging. However, the detection model operates as a **surveillance/classification system** and is not integrated into an autonomous trading architecture where a detected manipulation can directly prevent a trade.

Taken together, these studies reveal an important progression in AI-based trading research. Early multi-agent systems demonstrated the benefits of combining specialized market, prediction, sentiment, and decision agents [1], while modular reinforcement-learning systems demonstrated the value of reusable asset-specific agents [2]. More recent LLM-based systems have introduced increasingly sophisticated organizational structures, including analyst teams, debate mechanisms, risk-management teams, and portfolio managers [3], [4]. Risk-sensitive LLM/RL models have further demonstrated the importance of explicitly modelling risk [5], [6], while QuantHarness demonstrates the value of a dedicated risk-oriented agent for structured, price-driven trading decisions [7]. In parallel, cryptocurrency-manipulation research has established that pump-and-dump activity can be detected from real trading and communication data [8]–[10].

However, a common limitation remains across these research directions. **Specialized agents generally provide information, recommendations, risk scores, or votes to a downstream decision-maker, but their outputs do not constitute an independently enforced safety constraint on the final trade.** In particular, existing multi-agent trading frameworks do not sufficiently investigate a situation in which the primary trading agents receive a strongly bullish price-action signal during a coordinated manipulation event while an independently trained sentiment/anomaly agent identifies abnormal promotional behaviour.

This motivates the proposed research: an **independently trained Sentiment-Anomaly Agent with explicit veto authority over the trading decision**. Instead of treating sentiment merely as another feature, vote, or state variable, the proposed architecture separates the **profit-seeking trading pathway** from the **market-integrity/safety pathway**. The trading agents may generate a BUY/SELL/HOLD decision based on price, technical, and predictive signals, while the Sentiment-Anomaly Agent independently evaluates news and social signals for abnormal sentiment, coordinated promotion, hype, and potential manipulation. When the anomaly confidence exceeds a predefined threshold, the agent can **veto the proposed trade**, preventing execution regardless of the bullish recommendation produced by the trading agents.

The research novelty therefore lies not simply in adding another sentiment model, but in investigating whether **independent decision authority improves the robustness of an agentic trading system against adversarial market conditions**. The proposed system can be evaluated through a controlled comparison of **veto-enabled and veto-disabled configurations**, analogous to the ablation philosophy used in MSPM, while using real documented P&D events as manipulation test cases. The evaluation can measure not only conventional trading metrics such as cumulative return, Sharpe ratio, maximum drawdown, and hit rate, but also **manipulation avoidance rate, false-veto rate, prevented-loss rate, detection latency, and performance specifically during documented P&D events**. This creates a direct experimental connection between multi-agent trading research and cryptocurrency-manipulation research that is largely absent from the existing literature.

Keywords: *Multi-Agent Reinforcement Learning, Large Language Models, Financial Trading, FinBERT, Sentiment-Anomaly Detection, Cryptocurrency Market Manipulation, Pump-and-Dump Detection, Risk Management, Trading Agents, Veto Mechanism, Override Mechanism, Algorithmic Trading*

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